

Calibrated Perplexity for Robust AI-Generated Text Detection

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Problem: Domain Shift in Perplexity

Same text, different perplexity across domains

Example: A news article vs a scientific paper

- ▶ News model: perplexity = 11.7
- ▶ Science model: perplexity = 27.3

Single threshold fails across genres.

Research question: How to make perplexity-based detection robust to domain shift?

Related Work

- ▶ GLTR: Statistical detection using top-k rank analysis
- ▶ Binoculars: Zero-shot detection using cross-perplexity
- ▶ Chen et al.: Smoothing techniques for n-gram LMs
- ▶ Bakhteev et al.: Domain shift and length impact
- ▶ Gritsay et al.: Fragment detection in academic texts

No systematic approach to calibrate perplexity across different domains.

Problem Statement

Key observation: For the same text x , perplexity varies across domains:

$$\text{PPL}_{\mathcal{M}_{\text{news}}}(x) \neq \text{PPL}_{\mathcal{M}_{\text{science}}}(x)$$

Our goal: Make perplexity domain-invariant through calibration.

Minimization problem:

$$\sum_{i=1}^{N_d} \sum_{x_j \in X_i} (M_{\mu_i, i}(x) - y_{ij})^2 + \sum_{i < j \leq N_d} (\text{TH}_{M_{\mu_i, i}}(x) - \text{TH}_{M_{\mu_j, j}}(x))^2 \rightarrow \min_{\bar{\mu}}$$

We minimize the variance of thresholds in ensemble across different domains for the same text x , making the score domain-invariant.

Our Solution: Ensemble of Domain-Specific Models

Key Idea:

1. Train separate KenLM models on each domain
2. For a new text, compute perplexities from all models
3. Combine predictions via weighted voting

Advantages:

- ▶ No need to know the test text's domain
- ▶ Each model specializes in its domain
- ▶ Weighted voting balances human/AI detection

Training:

- ▶ Train trigram KenLM models on human texts from 7 domains
- ▶ Domains: news, hard_essays, scientific texts, stories, articles, hard_news, hard_science

Threshold Calibration:

$$\tau_d = \arg \max_{\tau} \text{F1}(\mathbf{1}[\text{PPL}_{\mathcal{M}_d}(x) < \tau], y(x))$$

Ensemble Decision:

$$H(x) = \mathbf{1} \left[\sum_{d \in \mathcal{D}} w_d \cdot \mathbf{1}[p_d < \tau_d] \geq 0.5 \right],$$

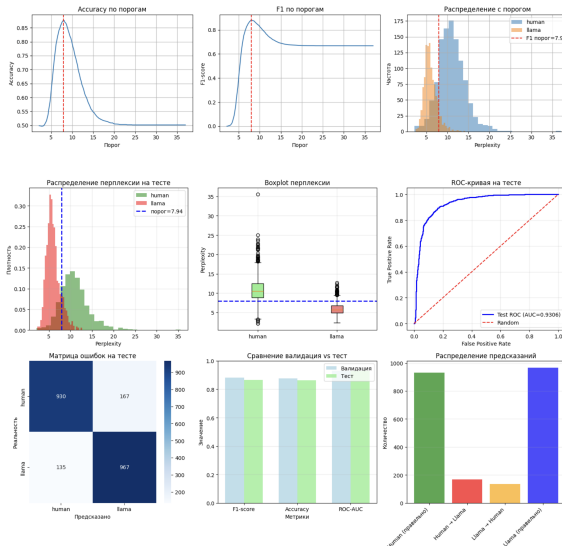
Experiment 1: Cross-Domain Perplexity Matrix

Train/Test	AINL	stories	reviews	articles
AINL (science)	11.6 / 6.9	27.3 / 23.4	26.1 / 21.9	23.8 / 19.2
stories	26.8 / 22.7	16.8 / 10.6	16.1 / 14.2	20.0 / 18.7
reviews	23.3 / 18.0	18.5 / 13.8	11.1 / 9.3	18.8 / 15.8
articles	19.9 / 14.2	20.1 / 15.2	17.7 / 13.6	13.3 / 9.0

Cell format: Human PPL / AI PPL. **Bold diagonal** = in-domain performance.

Observation: Perplexity increases significantly when moving off-diagonal, confirming domain shift.

Additional Visualizations



Experiment 2: Tokenization and Pruning

Goal: Improve model stability across domains

Methods tested:

- ▶ SentencePiece tokenization (vocab_size=8000)
- ▶ N-gram pruning (-prune 2 2 2)
- ▶ Size cutting
- ▶ Combined optimization

Result: Only 1–2% improvement – helpful but not sufficient.

Experiment 3: Ensemble Strategies

We compared four ensemble strategies:

Results:

Method	Human recall	AI recall
Simple majority	81%	71%
Perplexity-weighted	82%	70%
Percentage-weighted	81%	70%

Key insight:

- ▶ All ensembles show almost equal results

Conclusion

Key Contributions

- ▶ Demonstrated that perplexity thresholds are highly domain-dependent (vary by 2–3x)
- ▶ Proposed ensemble of domain-specific KenLM models for domain-agnostic detection
- ▶ Showed that ensemble voting achieves the most robust results

Future Work

- ▶ Upgrade combined ensemble
- ▶ Test on more diverse domains and languages
- ▶ Come to clusterization
- ▶ Combine and compare with other detectors